

TerraState: A Testable Predictive-State World Model for Weather-Driven Land-Surface Forecasting

Anonymous submission

Abstract

High-resolution satellite time series are a primary tool for monitoring vegetation, agriculture, and ecosystem response. Forecasting from these series is increasingly formulated as a weather-driven task: predicting future land-surface observations from cloud-obscured image histories and meteorological drivers. Yet such models are primarily evaluated by fixed-horizon pixel accuracy, which cannot establish whether an internal representation functions as a forecast-bearing, weather-responsive predictive state. An accurate forecaster may still ignore the weather forcing, collapse toward persistence, or expose a latent state that does not actually carry the forecast—failures that standard error metrics cannot detect. We introduce **TerraState**, a testable predictive-state world model. TerraState infers a spatial predictive state from cloud-masked histories. A shared transition advances this state under future weather, geography, and elapsed time, and a state readout converts the advanced state into an explicit contribution to the final forecast. Rather than treating architecture alone as evidence that a world state exists, TerraState makes its predictive-state claim falsifiable through state-contribution removal, a supporting identity-transition control, and matched interventions comparing actual future weather with matched-donor and normalized-mean weather. On GreenEarthNet under temporal distribution shift, TerraState retains useful forecasting skill; state removal degrades validation and OOD-t performance, and actual weather yields lower complete-window loss than both controls on a frozen heat–drought subset.

1 Introduction

High-resolution Earth-observation (EO) time series support localized monitoring of vegetation, agriculture, and ecosystem response. Forecasting future land-surface observations requires combining sparse, cloud-obscured satellite histories with past meteorology, static geography, and supplied future weather. EarthNet2021 formalized this setting as guided video prediction from past Sentinel-2 observations, topography, and future meteorology (Requena-Mesa et al. 2021); GreenEarthNet refined it with vegetation-focused cloud masking, temporal-distribution-shift evaluation, and high-resolution weather-conditioned forecasting

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(Benson et al. 2024). We study this task from a predictive-state world-modeling perspective: a model must summarize incomplete observations and represent how the observed surface may evolve under future forcing.

Weather-conditioned EO forecasters have improved the quality of predicted observations. Within standard EO forecasting benchmarks, however, the primary evidence remains output accuracy over a fixed forecast window (Requena-Mesa et al. 2021; Benson et al. 2024). Such accuracy shows whether outputs resemble observations, but cannot by itself establish whether an internal state contributes to those outputs or whether future meteorology advances that state. A predictor may bypass its latent state, use weather only weakly, or behave as a one-shot mapping; accurate outputs can also coexist with temporally disordered latent representations (Yang et al. 2026).

Predictive-state representations characterize state through future observables rather than an assumed hidden physical variable (Littman, Sutton, and Singh 2001). This motivates a concrete question: *can an EO forecaster expose a predictive state that both carries the forecast and mediates a faithful response to future weather?* Accordingly, we introduce TerraState, a testable predictive-state world model for weather-driven EO forecasting. TerraState makes these properties empirical claims rather than consequences of naming a hidden representation. Its scope is deliberately narrower than recovering a complete physical state or building a causal or general-purpose generative simulator.

Figure 1 summarizes the resulting model. TerraState infers a spatial predictive state from cloud-masked EO history, past meteorological observations, and static geography. A shared transition advances that state under supplied future weather, geography, and elapsed time. The state readout turns the advanced state into an explicit forecast contribution that is added to a context-only prediction, so state use is exposed in the final output rather than only in a hidden layer. During training, a frozen representation of observed future EO anchors the transitioned state without exposing future observations at inference. After training, state-contribution removal tests forecast relevance, while weather-path substitution tests the response to supplied forcing without retraining.

On GreenEarthNet OOD-t, TerraState obtains $R^2 = 0.569$ and $\text{RMSE} = 0.151$, retaining useful forecasting skill under temporal shift. Removing the state contribution reduces

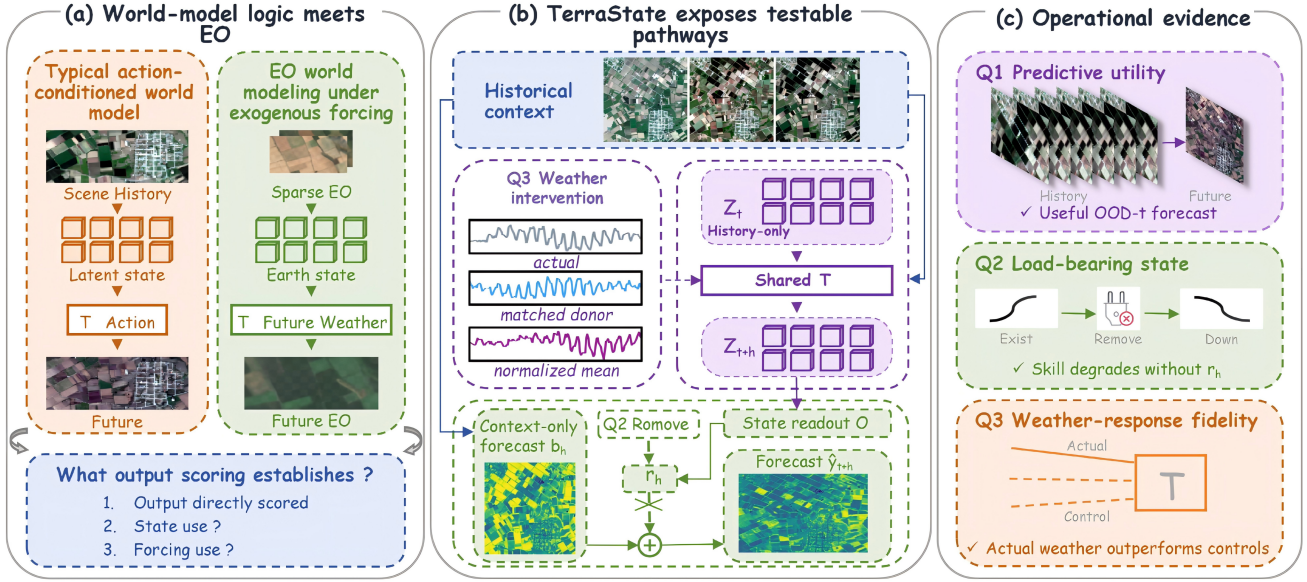


Figure 1: Conceptual overview of TerraState’s testable EO world-modeling contract. (a) Action-conditioned world-model logic maps to EO modeling under exogenous weather forcing, while output scoring alone leaves internal state and forcing use untested. (b) TerraState exposes a history-only predictive state, shared weather-conditioned transition, state readout, and intervention interfaces. (c) Q1–Q3 provide evidence for predictive utility, state contribution, and weather-response fidelity.

performance on both validation and OOD-t, with paired confidence intervals excluding zero. On the frozen matched subset, actual future weather yields lower masked loss over the complete 20-step forecast window than masked donor and normalized-mean controls. Together, these results support a load-bearing, weather-responsive predictive state under the evaluated protocol.

Our contributions are:

- We frame weather-driven EO world modeling around a falsifiable question: whether a history-derived predictive state both contributes to the forecast and mediates its response to future meteorological forcing.
- We introduce TerraState, which combines an explicit state-mediated forecast path, a shared weather-conditioned transition, and future-state anchoring from observed future EO during training.
- We evaluate the same trained model at three levels: useful temporal-shift forecasting, load-bearing state contribution on validation and OOD-t, and greater complete-window fidelity under actual weather than under matched-donor and normalized-mean controls.

2 Related Work

Weather-conditioned EO forecasting. Weather-conditioned EO forecasting predicts future land-surface observations from satellite histories, meteorology, and geographic context. EarthNet2021 formalized this guided video-prediction setting, and GreenEarthNet/Contextformer refined it for vegetation dynamics and temporal shift (Requena-Mesa et al. 2021; Benson et al. 2024). Deterministic methods use recurrent, convolutional, or transformer

predictors (Shi et al. 2015; Wang et al. 2017; Gao et al. 2022b,a), whereas video-diffusion models represent multiple plausible futures (Voleti, Jolicoeur-Martineau, and Pal 2022; Zhao et al. 2025). ViT-Koop advances a compressed EO state, and prior weather-response analysis perturbs meteorological inputs at the output level (Shinohara and Saomoto 2025; Diaconu et al. 2022). Across these strands, evidence centers on forecast outputs, with some studies also analyzing weather response or learned representations. This leaves a narrower question about whether an explicit internal state participates in the prediction path and responds to supplied weather.

World models: latent dynamics to interactive environments. World-model research supplies the broader context for this shift from output prediction to explicit internal state. A control-oriented lineage compresses observations and learns latent transitions for rollout, planning, or imagination (Ha and Schmidhuber 2018; Hafner et al. 2019, 2020). MuZero predicts planning-relevant policy, value, and reward (Schrittwieser et al. 2020). IRIS learns an agent inside a tokenized world model, whereas Genie learns action-controllable environments from video (Micheli, Alonso, and Fleuret 2023; Bruce et al. 2024). Drive-OccWorld connects action-conditioned occupancy forecasting to driving planning (Yang et al. 2025). These examples share a state-transition-prediction structure but optimize different downstream objectives, motivating a task-specific account of predictive state in EO.

EO world models and forcing-conditioned simulation. In EO, this shared structure must be specialized to partially observed geospatial processes under external environ-

mental drivers. Recent preprints make this connection explicit. EO-WM structures weather forcing for probabilistic EO forecasting and output-response diagnostics (Luo et al. 2026). VegSim rolls a latent vegetation state under user-specified weather for scenario-conditioned simulation (Iele et al. 2026). A cloud-aware model instead predicts future observation availability rather than land-surface pixels (Albughdadi 2026). Here, future weather is an exogenous input for forecasting EO observations rather than the prediction target itself. Complementing these objectives, TerraState examines whether the state used by a weather-conditioned EO forecast makes a removable contribution and whether actual forcing yields greater complete-window fidelity than frozen controls.

Predictive states and testability. Predictive-state work asks how internal state is defined, supervised, and evaluated. Classical predictive-state representations define state through future observables, and Predictive-State Decoders explicitly supervise recurrent states to predict those observables (Littman, Sutton, and Singh 2001; Venkatraman et al. 2017). I-JEPA and V-JEPA learn predictive representations without reconstructing raw pixels (Assran et al. 2023; Bardes et al. 2024). LatentTSF shows that accurate forecasts can coexist with temporally disordered latents (Yang et al. 2026), while PLSM constrains action effects in a control setting (Saanum, Dayan, and Schulz 2024). In automaton-governed generative-model settings, dedicated evaluation further reveals incoherence missed by standard diagnostics (Vafa et al. 2024). Together, these works motivate an EO-specific test: output accuracy alone cannot establish that an exposed state carries prediction or mediates weather forcing. Section 3 therefore constructs TerraState around an on-path predictive state, a shared weather-conditioned transition, and state-removal and weather-control interfaces that make this bounded claim testable.

3 Method

Problem Formulation and Model Overview

Weather-driven Earth-observation (EO) world modeling seeks to represent how the land surface evolves from incomplete satellite histories under future meteorological forcing. We formulate this task around a predictive state inferred from observation history, advanced by the supplied meteorological forcing, and decoded into future observations.

Accordingly, we introduce TerraState as a testable predictive-state world model. TerraState differs from a conventional EO predictor by making the predictive state’s contribution explicit in the final forecast. A history operator and projector construct z_t , and a shared transition advances it under the ordered future-weather prefix, static geography, and forecast horizon. A state readout decodes the advanced state into the state-mediated forecast contribution r_h . TerraState combines r_h with the context-only forecast b_h .

At forecast issue time t , let $\mathcal{H}_t = \{(x_i, m_i, \tau_i)\}_{i=1}^C$ denote cloud-obscured optical history with validity mask m_i and acquisition time τ_i . Historical weather is $u_{\leq t}^{\text{past}}$, and the ordered H -step future sequence is $u_{t+1:t+H}$; g denotes static geo-

graphy and $y_{t+1:t+H}$ the targets. Let $\tilde{\mathcal{C}}_t = (\mathcal{H}_t, u_{\leq t}^{\text{past}}, g)$ and query $h \in \{1, \dots, H\}$. TerraState realizes this formulation through:

$$\begin{aligned} (b_{1:H}, e_t) &= q_\theta(\tilde{\mathcal{C}}_t), \\ z_t &= P_\rho(e_t), \\ z_{t+h} &= T_\psi(z_t, u_{t+1:t+h}, g, h), \\ r_h &= O_\omega(z_{t+h}), \\ \hat{y}_{t+h} &= b_h + r_h. \end{aligned} \quad (1)$$

Here, q_θ emits context-only forecasts $b_{1:H}$ and spatial tokens e_t , P_ρ constructs z_t , and T_ψ advances it to z_{t+h} . The readout O_ω decodes the horizon-specific spatial forecast contribution r_h from the advanced state.

By construction, q_θ has access only to observation history, past weather, and static geography; it receives neither future EO observations nor future meteorological forcing. The future-weather sequence enters only through T_ψ , which directly advances the same z_t for each h . The explicit addition of r_h supports separate tests of whether the state carries predictive information and whether the forecast responds to the supplied weather. This formulation treats z_t as a predictive representation used by the forecast, rather than claiming recovery of a complete physical state or a causal simulator.

TerraState Architecture

This subsection details the inference architecture that realizes Equation 1. The history module produces a context-only forecast and a spatial predictive state. A shared transition advances that state under future meteorological forcing, and a state readout converts the advanced state into an additive forecast contribution. Figure 2 illustrates these three modules.

Historical Context and Spatial Predictive State. Historical context must support both the context-only forecast and the state that future forcing advances. In one forward pass, the history operator q_θ processes cloud-masked EO history, past meteorological observations, and static geography, and outputs both $b_{1:H}$ and spatial context tokens e_t . The projector maps the tokens at the final historical time to $z_t = P_\rho(e_t) \in \mathbb{R}^{N \times d}$. Each of the N tokens corresponds to a spatial patch, preserving the patch-level organization required by the later state readout. The context-only forecast and predictive state are therefore derived from the same historical context but remain distinct outputs. We instantiate q_θ with a pretrained PVT v2/Contextformer backbone (Wang et al. 2022; Benson et al. 2024).

Shared Weather-Conditioned Transition. Future weather conditions the predictive-state transition through its ordered prefix. A shared GRU weather encoder summarizes each prefix, while separate encoders represent patch-wise static geography and the queried horizon. For spatial patch $i \in \{1, \dots, N\}$,

$$d_h = E_u(u_{t+1:t+h}), \quad c_{h,i} = F([d_h; E_g(g)_i; E_h(h)]). \quad (2)$$

The weather code d_h and horizon embedding $E_h(h)$ are broadcast across patches, whereas $E_g(g)_i$ retains patch-wise

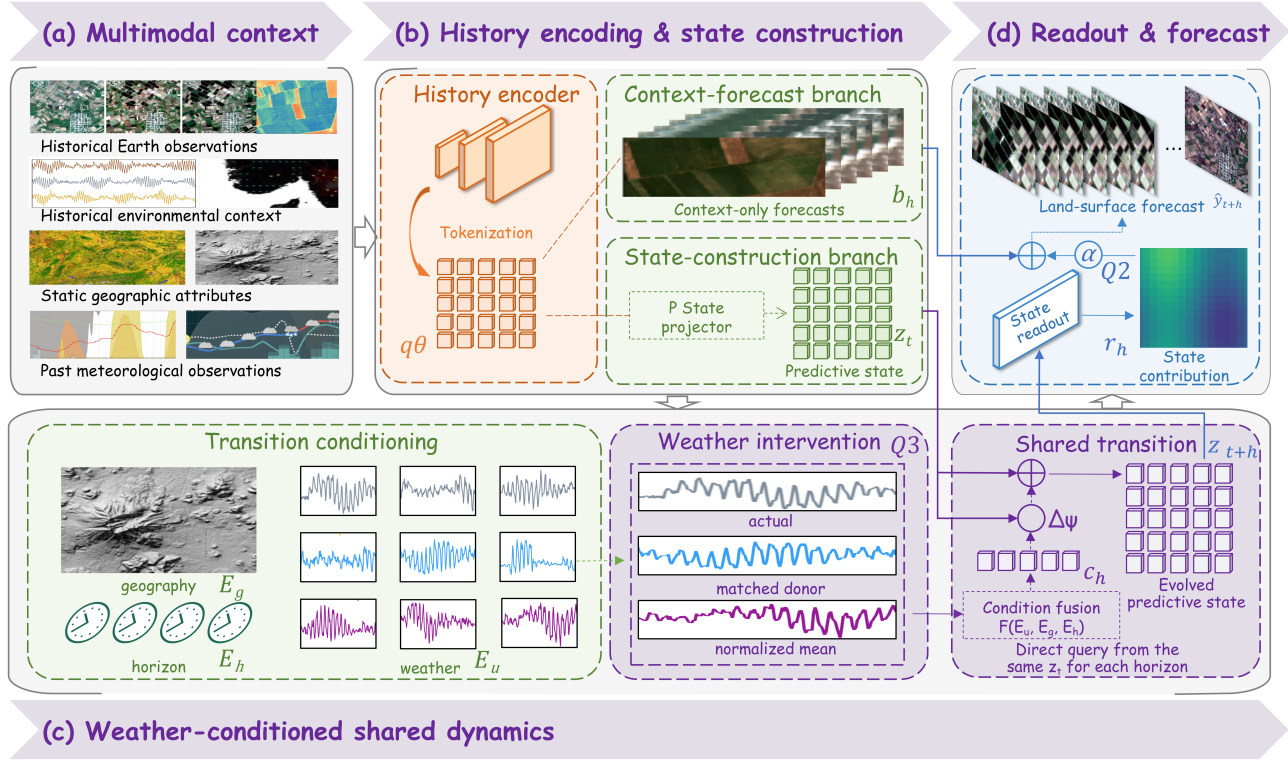


Figure 2: TerraState architecture and intervention interfaces. The diagram organizes multimodal historical context, history encoding and predictive-state construction, a shared weather-conditioned transition, and state readout into the final forecast. Cloud-masked EO history, past meteorological observations, and static geography form the context for the history encoder; future meteorological forcing enters the shared transition together with geography and the queried horizon. The transitioned state is read out as an explicit state contribution and combined with the context-only forecast. Q2 removes this state contribution, while Q3 holds the historical context fixed and replaces actual future weather with season-, geography-, and quality-matched donor weather or normalized-mean weather. These interventions test state contribution and forecast-window response fidelity, not composition or causal effects.

geographic variation. The resulting condition therefore varies with both horizon and spatial patch. It drives the residual state update

$$z_{t+h,i} = z_{t,i} + \Delta\psi([\text{LN}(z_{t,i}); c_{h,i}]). \quad (3)$$

The weather encoder, geography encoder, condition-fusion network, and transition parameters are shared across patches and horizons; the condition values themselves need not be spatially identical. For each queried h , the model applies the residual transition once to the same z_t , using $u_{t+1:t+h}$ and the corresponding horizon code. It does not recursively roll out $z_t \rightarrow z_{t+1} \rightarrow \dots$. Separating historical state construction from future forcing isolates future meteorological forcing within state evolution.

State Readout and Additive Forecast. To connect the transitioned state to observable predictions, the state readout O_ω maps each spatial token to a local 4×4 patch and reassembles the patches into the raster contribution r_h . This contribution is added to the context-only forecast:

$$r_h = O_\omega(z_{t+h}), \quad \hat{y}_{t+h} = b_h + \alpha r_h, \quad \alpha \equiv 1. \quad (4)$$

The coefficient α remains fixed throughout training and inference. The additive design isolates a distinct state-mediated forecast contribution for direct evaluation, while retaining the context-only forecast as its matched reference.

Future-Anchored State Learning

Training Identities and Purpose. To align the transitioned state with observed future evidence without exposing future EO at inference, training separates the deployable TerraState student from two frozen reference branches. An exact full-model warm start from a forecasting precursor initializes the student, which follows the inference chain in Equation 1. An independent full-weather *KD teacher* reads the EO observation history, past weather, static geography, and the complete future-weather sequence, but no future EO, and produces a stopped forecast target. The *future-state target encoder* is a training-start frozen copy of the student’s q_θ and P_ρ and constructs the latent target described below. Both reference branches remain frozen, gradients update only student parameters enabled by the training schedule, and neither branch is invoked at inference.

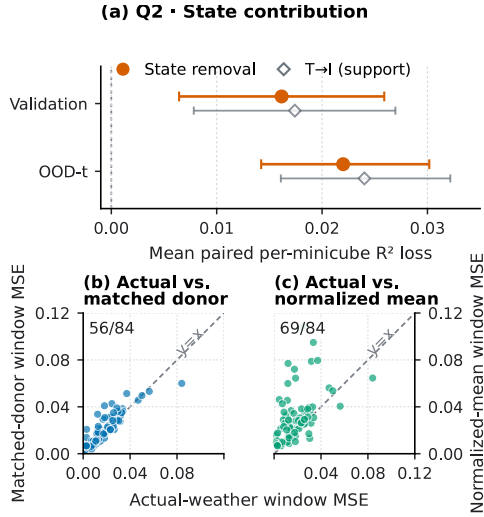


Figure 3: State and weather interventions. (a) Per-minicube ΔR^2 for state removal (filled, primary) and $T \rightarrow I$ (open, supporting), with paired-bootstrap 95% CIs. (b,c) Complete 20-step-window masked MSE for actual versus matched-donor and normalized-mean weather on 84 frozen pairs. Above-diagonal points favor actual weather; 56/84 and 69/84 are descriptive.

Forecast Objectives. Two forecast objectives train the student against observed NDVI and regularize it toward the full-weather teacher. Let b , h , and p index minicubes, forecast horizons, and raster pixels. We use c_{bhp} for the clear-observation indicator, v_{bp} for the vegetation indicator, and $a_{bp} = \max_h \mathbf{1}[\hat{y}_{bhp} \neq -1]$ for prediction validity. With $\hat{y}^{\text{tea}}$ denoting the teacher forecast,

$$\begin{aligned} \bar{\ell}_{bp}^{\text{GT}} &= \frac{\sum_h c_{bhp} (\hat{y}_{bhp} - y_{bhp})^2}{\sum_h c_{bhp} + \epsilon_{\text{pix}}}, \\ \mathcal{L}_{\text{GT}} &= \frac{\sum_{b,p} v_{bp} a_{bp} \bar{\ell}_{bp}^{\text{GT}}}{\sum_{b,p} v_{bp} a_{bp} + \epsilon_{\text{GT}}}, \\ \mathcal{L}_{\text{KD}} &= \frac{\sum_{b,h,p} c_{bhp} v_{bp} (\hat{y}_{bhp} - \text{sg}[\hat{y}_{bhp}^{\text{tea}}])^2}{\sum_{b,h,p} c_{bhp} v_{bp} + \epsilon_{\text{KD}}}. \end{aligned} \quad (5)$$

Thus, $\mathcal{L}_{\text{GT}}$ first normalizes each pixel by its clear horizons and then averages over vegetation pixels with valid predictions. $\mathcal{L}_{\text{KD}}$ instead forms one global mean over clear vegetation time-pixel elements; its stop-gradient target regularizes the student without updating the teacher.

Future-State Representation Target. The future-state objective anchors the terminal transitioned state to a representation constructed from observed future EO. Define $\mathcal{C}_{t+H}^*$ as the observed all-frame EO sequence with its recorded masks, including future frames, together with past weather and static geography; future weather is zeroed in this branch. The frozen copy runs the history operator in all-frames-visible encoding mode, with every temporal position supplied together with its recorded mask. It then selects the terminal spatial token and applies its frozen projector. Only

the terminal future-state target is formed. For terminal spatial patch i ,

$$\begin{aligned} z_{t+H,i}^* &= \text{sg}[P_{\rho^0}(q_{\theta^0}(\mathcal{C}_{t+H}^*)_{t+H,i})], \\ \ell_i &= 1 - \cos(\text{LN}(z_{t+H,i}), \text{LN}(z_{t+H,i}^*)), \\ \mathcal{L}_{\text{FS}} &= \frac{\sum_i m_i \ell_i}{\sum_i m_i + \epsilon_{\text{FS}}}, \end{aligned} \quad (6)$$

where (θ^0, ρ^0) denotes the training-start frozen copy and sg stops target gradients; i indexes terminal patches across the batch. The mask m_i requires the terminal 4×4 patch to be fully clear and to contain at least one vegetation pixel. Future EO is used only to construct a stopped, training-only target; it is never an input to the student forecast or inference graph.

Total Objective and Inference Boundary. The complete training objective is

$$\mathcal{L} = \mathcal{L}_{\text{GT}} + 0.5 \mathcal{L}_{\text{KD}} + \lambda_s \mathcal{L}_{\text{FS}}. \quad (7)$$

The KD teacher and future-state target encoder are discarded after training, leaving the student inference graph unchanged. Future-state alignment shapes what the transitioned state represents; the following interfaces test whether that state affects forecasts and responds to supplied weather.

Testable Predictive-State Interfaces

We therefore define two post-training interfaces on the same frozen TerraState model. Each interface changes only a designated part of the forward computation, requires no retraining, and introduces no additional objective.

State-contribution intervention. This interface tests whether the explicit state contribution improves the forecast. Immediately before the addition in Equation 4, we temporarily set $\alpha = 0$, yielding $\hat{y}_{t+h}^{\text{remove}} = b_h$. The frozen model, evaluated sample, historical context, state construction, transition, readout, and ground-truth forecast window remain fixed. We call the state-mediated contribution *load-bearing* when removing r_h degrades paired forecast quality in the expected direction and the prespecified uncertainty interval excludes zero. Section 4 specifies the forecast metric and uncertainty analysis. Replacing T_ψ by the identity is only a supporting diagnostic of transition involvement; it is not part of the load-bearing definition because the identity substitution presents O_ω with states outside its trained input distribution.

Controlled weather-path substitution. This interface tests whether future weather affects the forecast through the state-mediated path and whether that response has forecast-window fidelity. It fixes the frozen model, evaluated sample, historical context, $b_{1:H}$, z_t , static geography, queried horizon, readout, and ground-truth forecast window. Only the future-weather sequence supplied to T_ψ is substituted among actual weather, matched-donor weather, and normalized-mean weather:

$$\begin{aligned} \hat{y}_{t+h}(u) &= b_h + O_\omega(T_\psi(z_t, u_{t+1:t+h}, g, h)), \\ \Delta \mathcal{L}_{\text{ctrl}} &= \mathcal{L}_{\text{win}}(\hat{\mathbf{y}}(u^{\text{ctrl}}), \mathbf{y}) \\ &\quad - \mathcal{L}_{\text{win}}(\hat{\mathbf{y}}(u^{\text{act}}), \mathbf{y}), \end{aligned} \quad (8)$$

where $u \in \{u^{\text{act}}, u^{\text{don}}, u^{\text{mean}}\}$, $\text{ctrl} \in \{\text{don}, \text{mean}\}$, and $\mathcal{L}_{\text{win}}$ is the masked loss over the complete 20-step forecast window. A positive ΔL_{ctrl} means that actual weather yields lower error than the corresponding control. A response is *detectable* when the masked mean absolute forecast difference between actual and control weather, computed per minicube over the common forecast mask, is nonzero. Forecast-window response fidelity requires positive ΔL_{ctrl} separately against both frozen controls, with reliability determined by the prespecified uncertainty analysis in Section 4. We call the predictive state *weather-responsive* only when weather substitution through the state-mediated path produces a detectable forecast response and actual weather has positive fidelity against both controls. Section 4 specifies the control construction. This operation is a controlled diagnostic substitution; it does not estimate a causal effect, guarantee counterfactual correctness, or establish an extreme-specific enhancement.

4 Experiments

Experimental Setup

We evaluate one forecasting prerequisite and two internal properties of the predictive state. All three questions use the same final TerraState model after the complete 40-epoch, 14,880-update training protocol; Q2 and Q3 alter only its frozen forward computation and require no retraining:

1. **Q1—Forecasting performance:** Does TerraState retain useful forecasting performance under temporal shift (OOD-t)?
2. **Q2—State contribution:** Does removing the state-mediated forecast contribution reduce prediction quality?
3. **Q3—Weather-forcing response:** Does actual future weather predict the complete forecast window more faithfully than matched-donor and normalized-mean controls?

Dataset and protocol. GreenEarthNet provides the common evaluation setting (Benson et al. 2024). Each minicube contains 30 five-day Sentinel-2 composites at 128×128 pixels and 20 m ground sampling, together with aligned meteorology, cloud and quality masks, and static geography. The first 10 composites provide the historical context, and the following 20 form the prediction window. The temporal out-of-distribution split (OOD-t) contains 1,904 minicubes. OOD-t and the Q2–Q3 intervention results are held out from model selection.

Metrics and statistical units. The metrics distinguish dataset-level forecasting performance from paired and clustered intervention effects. The target is NDVI over valid vegetation pixels. Q1 reports R^2 , root mean squared error (RMSE), Nash–Sutcliffe efficiency (NSE), absolute prediction bias, and RMSE_{25} , the RMSE over the first 25 forecast days. For Q2, the official ΔR^2 is the difference between the full and intervened dataset-level scores. We report this estimand separately from the mean per-minicube paired ΔR^2 and its paired-bootstrap 95% confidence interval. Q3 uses masked mean squared error over the complete 20-step forecast window. Its primary uncertainty interval resamples

geographic clusters, preserving the dependence induced by matched controls.

Comparison purpose. Forecasting comparisons establish the performance context for Q1. We compare TerraState with non-learning, recurrent, video-prediction, and transformer-based forecasting methods in Table 1 (Benson et al. 2024). This profile tests whether TerraState retains useful forecast skill before its internal state is examined; load-bearing state use and weather response are determined by the Q2 and Q3 interventions rather than by table rank.

Implementation and model selection. TerraState contains 7.18M parameters. We train it with AdamW for 40 epochs (14,880 updates), using a global batch size of 64 and a core learning rate of 3×10^{-5} for the non- q branch. The final model used in Q1–Q3 completes this full training protocol and is selected solely by validation forecasting performance.

Forecasting Performance under Temporal Shift

TerraState retains useful forecasting skill on the GreenEarthNet OOD-t split under temporal distribution shift. Across 1,904 minicubes, it obtains $R^2 = 0.56935$ and $\text{RMSE} = 0.15059$ (Table 1). Its $\text{RMSE}_{25} = 0.082$ indicates low error over the first 25 forecast days and is the metric on which TerraState compares most favorably with the listed methods. The overall profile is mixed: $\text{RMSE} = 0.151$ lies within the numerical range of several learned forecasters, whereas its R^2 and NSE are not the largest values in the table. Q1 therefore establishes the forecasting prerequisite for the predictive-state analysis; Q2 and Q3 separately evaluate the same model’s state contribution and weather response through the controlled interventions reported in the following sections.

Load-Bearing Predictive State

Table 2 reports the primary state-removal intervention and the supporting $T_\psi \rightarrow I$ diagnostic on Validation and OOD-t. TerraState’s explicit state-mediated contribution is load-bearing on both splits, as established by state removal; $T_\psi \rightarrow I$ is retained only as a diagnostic of learned-transition involvement.

State removal yields paired mean ΔR^2 of 0.01616 on Validation (95% CI [0.00643, 0.02590], $n = 589$) and 0.02200 on OOD-t (95% CI [0.01422, 0.03018], $n = 1,019$); both paired-bootstrap intervals exclude zero. On the dataset-level scorer, the corresponding full-minus-removal differences are 0.01121 and 0.01997. Together, these effects show that the explicit state path carries a measurable forecast increment across both splits, without implying that all predictive information passes through it. Replacing the learned transition with the identity produces degradation in the same direction, supporting transition involvement. Because identity substitution may present the readout with states outside its trained input distribution, it does not establish transition necessity.

Weather-Forcing Response

TerraState’s state-mediated path responds detectably to supplied future weather, while actual weather achieves greater

Method	$R^2 \uparrow$	RMSE $\downarrow$	NSE $\uparrow$	Bias $\downarrow$	RMSE ₂₅ $\downarrow$	#Params
Persistence	0.000	0.230	-1.280	0.170	0.090	0
Previous year	0.560	0.200	-0.400	0.140	0.180	0
Climatology	0.580	0.180	-0.340	0.130	0.160	0
ConvLSTM	0.580	0.160	-0.130	0.110	0.110	1.0M
Earthformer	0.520	0.160	-0.130	0.100	0.090	60.6M
PredRNN	0.620	0.150	0.030	0.100	0.100	1.4M
SimVP	0.600	0.150	0.030	0.090	0.100	6.6M
Contextformer	0.620	0.140	0.090	0.090	0.080	6.1M
TerraState	0.569	0.151	-0.099	0.101	0.082	7.18M

Table 1: GreenEarthNet temporal-shift forecasting. RMSE₂₅ covers the first 25 forecast days; RMSE, absolute bias, and RMSE₂₅ are lower-is-better.

Split	Configuration	$R^2 \uparrow$	RMSE $\downarrow$	Official $\Delta R^2 \uparrow$	Paired ΔR^2 [95% CI]
Validation	Full TerraState	0.49732	0.15729	reference	—
	State removed	0.48611	0.17101	0.01121	0.01616 [0.00643, 0.02590]
	$T_\psi = \text{Id}$	0.48542	0.26102	0.01191	0.01742 [0.00782, 0.02696]
OOD-t	Full TerraState	0.56935	0.15059	reference	—
	State removed	0.54938	0.16519	0.01997	0.02200 [0.01422, 0.03018]
	$T_\psi = \text{Id}$	0.54766	0.25832	0.02169	0.02402 [0.01609, 0.03217]

Table 2: Q2 interventions. Official ΔR^2 is dataset-level; paired ΔR^2 is the per-minicube mean with paired-bootstrap 95% CI (Validation $n = 589$; OOD-t $n = 1,019$). State removal is primary; $T_\psi \rightarrow I$ is supporting.

Future weather	$R^2 \uparrow$	RMSE $\downarrow$	ΔLoss [95% CI] $\uparrow$	Actual lower
Actual	0.6254	0.1492	reference	—
Matched donor	0.5893	0.1584	0.00257 [0.00112, 0.00399]	56/84
Normalized mean	0.5430	0.1971	0.01126 [0.00547, 0.01708]	69/84

Table 3: Weather interventions on 84 frozen matched pairs. ΔLoss is the masked loss over the complete 20-step forecast window, computed as control minus actual (positive values favor actual); intervals are geographic-cluster 95% CIs and counts are descriptive. R^2 and RMSE apply only to the matched subset.

complete-window predictive fidelity than both frozen controls under the matched protocol. Across 84 frozen matched pairs from the predeclared extreme-weather stratum, history, initial state, geography, horizon, readout, sample, mask, and the ground-truth window remain fixed; only future weather entering the transition changes. Matched-donor weather is season-, geography-, and quality-matched; normalized-mean weather is zero in the frozen global z-score space. Figure 3(b,c) shows per-pair distributions; Table 3 reports effects and scores.

The per-minicube masked mean absolute forecast difference over the common forecast mask is 0.03592 for actual versus matched-donor weather and 0.08137 for actual versus normalized-mean weather; all 84 pairwise values are finite and positive. For the complete 20-step window, control-minus-actual ΔLoss is 0.00257 for matched donor (geographic-cluster 95% CI [0.00112, 0.00399]) and 0.01126 for normalized mean (95% CI [0.00547, 0.01708]); both intervals exclude zero. Under the frozen matched protocol, ac-

tual weather therefore has greater complete-window fidelity; together with Q2, this supports a forecast-bearing, weather-responsive predictive state.

5 Limitations and Scope

TerraState learns a future-predictive representation from satellite history and supplied meteorology, not a complete physical land-surface state. Its transitions use realized future weather; deployment with forecast meteorology may introduce unmeasured distribution shift.

Weather controls establish conditional predictive fidelity, not causal or counterfactual validity. The hot-dry interval does not support extreme-specific enhancement, and state removal isolates a measurable state-mediated increment without implying that all information passes through this state.

Evaluation is limited to GreenEarthNet temporal shift. Cloud screening and unobserved soil moisture, irrigation, and vegetation type may also limit the learned dynamics.

6 Conclusion

TerraState makes internal predictive-state claims in weather-driven EO forecasting empirically testable. It combines a history-derived spatial state, a shared weather-conditioned transition, an explicit forecast contribution, and future-state anchoring with post-training state-removal and weather-substitution tests. Under the evaluated protocol, TerraState retains useful OOD-t skill, forecast performance degrades after state removal, and actual weather gives greater complete-window fidelity than frozen controls. These results support a forecast-bearing, weather-responsive predictive state without establishing a complete physical or causal world model.

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